

How to solve a linear system?

$$Ax = b$$

e.g.
$$\begin{pmatrix} 2 & 4 & -2 \\ x & 9 & -3 \\ -2 & -3 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 10 \end{pmatrix}$$

Idea: do Gaussian elimination first so the equation becomes

$$\begin{pmatrix} 2 & 4 & -2 \\ & 1 & 1 \\ & & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 8 \end{pmatrix}$$

then can do backward substitution to solve x_1, x_2, x_3 easily.

$$4x_3 = 8 \Rightarrow x_3 = 2$$

$$x_2 + x_3 = 8 \Rightarrow x_2 = 6$$

$$2x_1 + 4x_2 - 2x_3 = 2 \Rightarrow x_1 = -1$$

Gaussian elimination

$$\begin{pmatrix} 2 & 4 & -2 & 2 \\ x & 9 & -3 & 8 \\ -2 & -3 & 7 & 10 \end{pmatrix} \xrightarrow[\text{2} \times \text{first row}]{\text{second row} -} \begin{pmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 6 \\ -2 & -3 & 7 & 10 \end{pmatrix}$$

$$\xrightarrow[\text{first row}]{\text{third row} +} \begin{pmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 6 \\ 0 & 1 & 5 & 12 \end{pmatrix}$$

$$\xrightarrow[\text{second row}]{\text{third row} -} \begin{pmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 4 & 6 \end{pmatrix}$$

Each elimination step has a simple matrix representation

$$\text{second row} - 2 \times \text{first row} = E_{21} = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

To generalize results, we have

$$\underbrace{E_{m, m-1} \cdots E_{m, 2} E_{m, 1}}_{\substack{\text{elimination} \\ \text{steps}}} \cdots \underbrace{E_{m-1, m-1} \cdots E_{m-1, 2} E_{m-1, 1}}_{\substack{\text{elimination} \\ \text{steps}}} A = U \quad \rightarrow \text{upper triangular matrix}$$

$$\Rightarrow A = E_{m, m-1}^{-1} \cdots E_{m, 1}^{-1} \cdots E_{m-1, m-1}^{-1} \cdots E_{m-1, 1}^{-1} U$$

Result 1:

if $E_{ij} = \begin{pmatrix} 1 & & \\ & l_{ij} & \\ & & \ddots \\ & & & 1 \end{pmatrix}$ (not entry 'i,j')

then $E_{ij}^{-1} = \begin{pmatrix} 1 & & \\ & -l_{ij} & \\ & & \ddots \\ & & & 1 \end{pmatrix}$ (not entry 'i,j')

Result 2: the result hold because of the order of operations

$$\underbrace{E_{m, m-1}^{-1} \cdots E_{m, 1}^{-1} \cdots E_{m-1, m-1}^{-1} \cdots E_{m-1, 1}^{-1}}_{\substack{\text{elimination} \\ \text{steps}}} = \begin{pmatrix} 1 & & & \\ -l_{1,2} & 1 & & \\ -l_{1,3} & -l_{2,3} & 1 & \\ \vdots & \vdots & \vdots & \ddots \\ -l_{1,m} & -l_{2,m} & \cdots & 1 \end{pmatrix}$$

We can prove this by induction

$$E_{ij}^{-1} (E_{ij} \cdots E_{m, m-1}) = \begin{pmatrix} 1 & & \\ & -l_{ij} & \\ & & \ddots \\ & & & 1 \end{pmatrix} \begin{pmatrix} 1 & & \\ & 0 & \\ & & \ddots \\ & & & 1 \end{pmatrix} = \begin{pmatrix} 1 & & \\ & -l_{ij} & \\ & & \ddots \\ & & & 1 \end{pmatrix}$$

(not entry 'i,j')

these could be non-zero

the effect of this matrix: Gaussian elimination step

it takes i -th row, times l_{ij} and then add it to j -th row where $j > i$. (So this operation would effect only entry 'i,j')

So we have $A = LU$ this is known LU decomposition

LU decomposition is good, but it is not always possible.
For example, if

$$A = \begin{pmatrix} 0 & 2 & 3 \\ 1 & 3 & 5 \\ 4 & 0 & 2 \end{pmatrix} \rightarrow \text{in this case, the first row cannot eliminate the second or third row}$$

Row exchange matrix $P_{ij} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} \rightarrow \text{identity matrix swap } i\text{-th and } j\text{-th row}$

$P_{ij} A$
effect of this matrix. Swap A_i and A_j .

important property of P_{ij}

$$P_{ij} = P_{ij}^{-1}$$

So in many cases we have

$$(E \dots P E \dots P E \dots) A = U$$

$$\Rightarrow (E' \dots E' \dots E') (P \dots P) A = U$$

$$\Rightarrow PA = LU$$

this is obvious because

$$P_{ij}^{-1} = I$$

$\rightarrow$ swap i and j
swap i and j again

$$\Rightarrow PE = E'P$$

$$\text{where } E' = PEP$$

if E is a elimination step, we have

$$\begin{aligned} & E_{32} P_{13} E_{31} E_{21} \\ &= E_{32} E'_{31} P_{13} E_{21} \\ &= E_{32} E'_{31} E_{21} P_{13} \end{aligned}$$

where $E'_{31} = P_{23} E_{31} P_{23}$
 $= P_{23} (I + L_{31} e_3^T e_1) P_{23}$

$$= I + L_{31} P_{23} e_3^T e_1 P_{23}$$

$$= I + \underline{L_{31} e_3^T e_1} \rightarrow \text{goes to row 2}$$

Similarly $E'_{21} = I + \underline{L_{21} e_2^T e_1} \rightarrow \text{goes to row 3}$